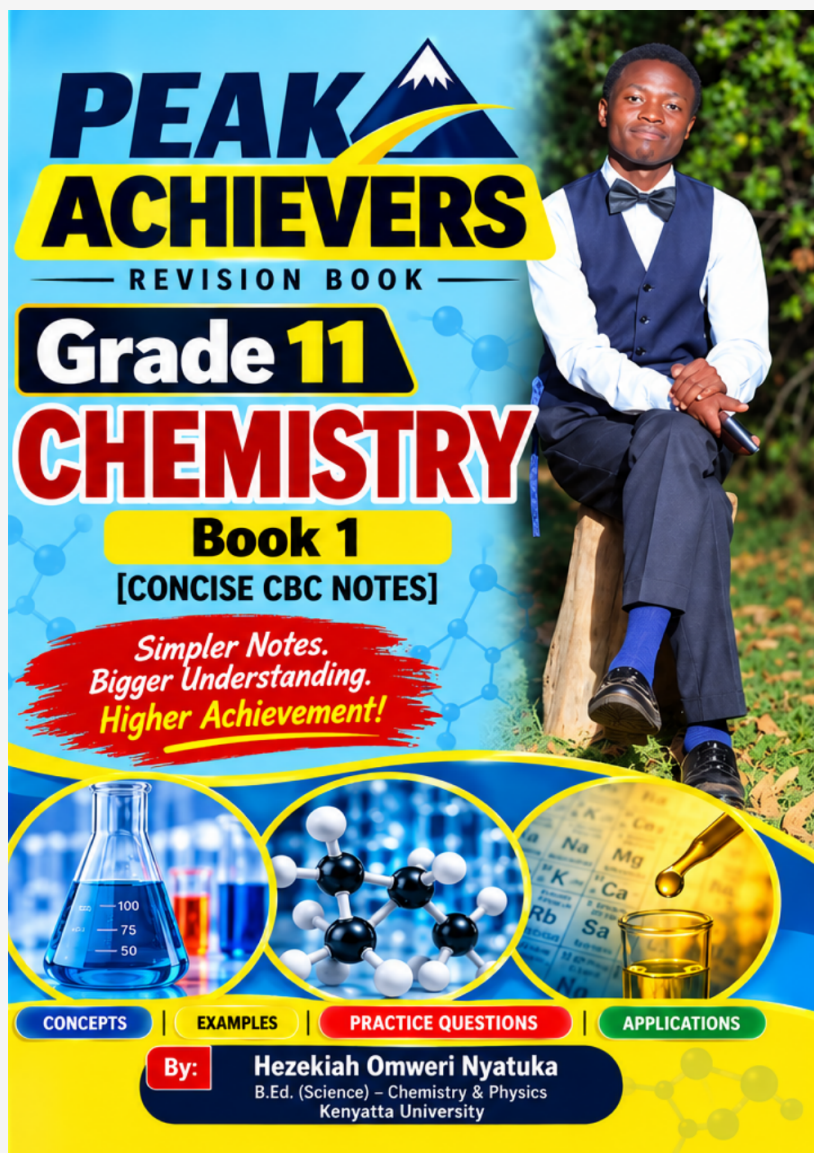


PEAK ACHIEVERS

GRADE 11 SENIOR SCHOOL CHEMISTRY

FREE SAMPLE PACK - BOOK 1

CONCISE NOTES



FULL BOOK PRICE
KSh 450

This sample includes:

- selected notes/pages
- worked examples & questions
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PEAK ACHIEVERS

GRADE 11 SENIOR SCHOOL CHEMISTRY

BOOK 1: CONCISE NOTES

Competency-Based Education (CBE)

Chemistry made clear • practical • applicable • achievable

For Grade 11 Senior School Learners

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Dedication

— **Hezekiah Omweri Nyatuka**

To each of you, this book carries a part of your contribution to my journey.

And to every learner who opens this book: may your curiosity grow, your confidence strengthen and your effort lead you towards excellence.

To my fellow teachers, whose commitment to education continues to inspire the work of helping learners succeed.

To Her, my special one, for being a special part of my journey and for the place you hold in my heart.

To my friends, Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani, and many others, whose friendship and encouragement have been part of my journey.

To my guardians, Peter Mungai Mong'are and Teresa Kwamboka Sawe, for their guidance, encouragement and support.

To my mother, Anne Kerubo, for her love, strength and unwavering support throughout my life.

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifices and belief in education continue to inspire my journey and the work I do as an educator.

Preface

Peak Achievers Grade 11 Senior School Chemistry Book 1 presents Chemistry in a clear, concise and application-oriented way. It combines key concepts, equations, worked examples, practical and inquiry activities, real-life applications, environmental awareness and focused practice.

Learning pathway: LEARN → UNDERSTAND → OBSERVE → APPLY → PRACTISE → ACHIEVE

How to Use This Book

Read the concise explanation first, study the key equation or sketch, follow the worked example, connect the idea to a practical or real-life situation, and then attempt the practice questions.

Feature Key

REMEMBER - essential fact

HINT - problem-solving guidance

NOTE / COMMON ERROR - warning or misconception

WORKED EXAMPLE - guided calculation

PRACTICE - short learner task

PRACTICAL / INQUIRY - investigation

APPLICATION - real-life or environmental connection

THINK BIG - higher-order reasoning

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CHAPTER 1 - THE MOLE

Learning Focus

Use the mole concept to describe quantities of substances, relate moles to mass and particles, perform chemical calculations, prepare solutions of known concentration and apply quantitative Chemistry in practical situations.

1.1 Mole Concept

A mole is the amount of substance containing 6.022×10^{23} specified particles. The Avogadro constant is $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$. The particles may be atoms, molecules or ions.

1.2 Moles and Particles

 **REMEMBER:** Always identify the particle named in the question: atoms, molecules or ions.

 **WORKED EXAMPLE:** 0.50 mol CO_2 contains $N = 0.50 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$ molecules.

 **PRACTICE:** Calculate the number of molecules in 0.25 mol H_2O .

$N = nN_A$ and $n = N/N_A$. Example: 0.50 mol CO_2 contains 3.011×10^{23} molecules.

1.3 Relative Atomic Mass

Relative atomic mass, A_r , is measured relative to 1/12 of the mass of a carbon-12 atom. It has no unit.

1.4 Relative Formula / Molecular Mass

Add the relative atomic masses of all atoms present. H_2O : $M_r = 18$. CaCO_3 : $M_r = 100$.

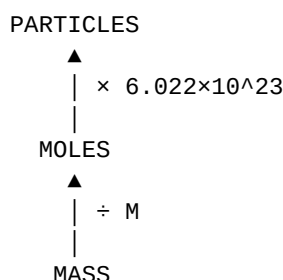
1.5 Molar Mass

 **HINT:** When mass is given, convert to moles first using $n = m/M$.

 **COMMON ERROR:** Do not confuse relative formula mass, which has no unit, with molar mass in g mol^{-1} .

$n = m/M$ and $m = nM$, where M is in g mol^{-1} . Example: $12 \text{ g Mg} \div 24 \text{ g mol}^{-1} = 0.50 \text{ mol}$.

1.6 Mole Conversion Map



Most quantitative problems become easier when the given quantity is first converted to moles.

1.7 Mole Ratios

A balanced equation gives reacting mole ratios. For $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$, $\text{H}_2:\text{O}_2:\text{H}_2\text{O} = 2:1:2$. Use coefficients, not subscripts.

1.8 Reacting Masses

 **WORKED EXAMPLE:** $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. 12 g Mg = $12/24 = 0.50$ mol. Ratio Mg:MgO = 1:1, so MgO = 0.50 mol. Mass = $0.50 \times 40 = 20$ g.

 **THINK BIG:** A factory uses more reactant than the balanced equation requires. How could this affect cost, waste and sustainability?

Route: mass given $\rightarrow$ moles $\rightarrow$ equation ratio $\rightarrow$ moles required $\rightarrow$ required mass. Example: 12 g Mg forms 20 g MgO in $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$.

1.9 Percentage Composition

% element = (mass contribution of element $\div$ formula mass) $\times 100$. % oxygen in $\text{H}_2\text{O} = 16/18 \times 100 = 88.9\%$.

1.10 Empirical Formula

Route: MASS $\rightarrow$ MOLES $\rightarrow$ DIVIDE BY SMALLEST $\rightarrow$ WHOLE-NUMBER RATIO. Example: 24 g C and 4 g H gives CH_2 .

1.11 Molecular Formula

Molecular formula = (empirical formula) $_n$, where $n = M_r \div$ empirical formula mass. CH_2 with M_r 28 gives C_2H_4 .

1.12 Concentration

$C = n/V$ and $n = CV$. Convert cm^3 to dm^3 : $1000 \text{ cm}^3 = 1 \text{ dm}^3$.

1.13 Preparing a Solution of Known Concentration

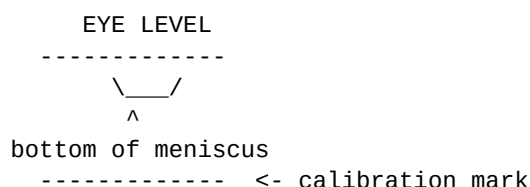
WEIGH $\rightarrow$ DISSOLVE $\rightarrow$ TRANSFER $\rightarrow$ RINSE

$\downarrow$
MAKE TO MARK $\rightarrow$ MIX

 **PRACTICAL / INQUIRY:** Prepare a teacher-approved solution of known concentration. Record the mass used, final volume and precautions taken.

 **HINT:** Read the bottom of the meniscus at eye level to avoid parallax error.

Read the bottom of the meniscus at eye level for a colourless aqueous solution.



1.14 Solution Stoichiometry

Use volume + concentration → moles → balanced-equation ratio → required quantity.

1.15 Application

Mole calculations support efficient use of raw materials, solution preparation, laboratory analysis and fertiliser manufacture, reducing waste and cost.

PEAK ACHIEVERS CHECKPOINT

Before moving on, explain the key idea in your own words, identify one real-life application and attempt the practice without looking back at the notes.

Quick Practice

1. Define a mole.
2. State the Avogadro constant.
3. Calculate moles in 16 g O₂.
4. Find molecules in 0.25 mol CO₂.
5. Determine molar mass of CaCO₃.
6. Calculate mass of 0.30 mol NaOH.
7. Explain the role of coefficients in a balanced equation.
8. Calculate concentration of 0.10 mol in 250 cm³.
9. Why is a meniscus read at eye level?
10. State one practical use of mole calculations.

CHAPTER 2 - NON-METALS

Learning Focus

Describe selected non-metals and their compounds, relate properties to uses, interpret important reactions and appreciate their agricultural, industrial and environmental significance.

2.1 General Properties

Non-metals commonly form covalent substances and show varied physical properties. Their chemical behaviour depends on electronic structure and bonding.

2.2 Nitrogen

Nitrogen, N_2 , forms about 78% of the atmosphere by volume. The $N \equiv N$ bond contributes to its relatively low reactivity under ordinary conditions.

2.3 Uses of Nitrogen

Uses include ammonia and fertiliser manufacture, inert atmospheres and food packaging.

2.4 Ammonia

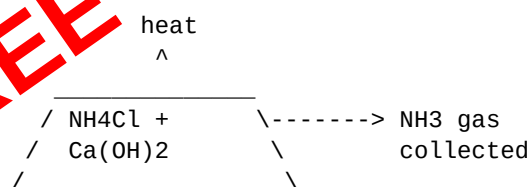
 **REMEMBER:** Ammonia is alkaline in water: damp red litmus turns blue.

 **APPLICATION:** Ammonia is a major starting material for nitrogen fertilisers.

NH_3 is colourless, pungent, highly soluble in water and forms an alkaline solution. Damp red litmus turns blue.

2.5 Laboratory Preparation of Ammonia

$2NH_4Cl(s) + Ca(OH)_2(s) \rightarrow CaCl_2(s) + 2NH_3(g) + 2H_2O(l)$. Ammonia is not normally collected over water because it is very soluble.



Note: NH_3 is not collected over water because it is highly soluble.

2.6 Test for Ammonia

$NH_3 + HCl \rightarrow NH_4Cl$ produces dense white fumes. Damp red litmus turns blue.

2.7 Manufacture of Ammonia

 **THINK BIG:** Why does an industrial plant recycle unreacted nitrogen and hydrogen instead of releasing them?

$N_2 + 3H_2 \rightleftharpoons 2NH_3$. An iron catalyst is used and unreacted gases can be recycled.

Quick Practice

1. Define a salt.
2. Name the salt family formed from nitric acid.
3. Describe preparation of copper(II) sulfate from copper(II) oxide.
4. State the carbon dioxide test.
5. Write the AgCl ionic equation.
6. Explain crystallisation.
7. What is water of crystallisation?
8. Why heat to constant mass?
9. State the colour change when hydrated copper(II) sulfate is strongly heated.
10. Explain one environmental effect of excessive fertiliser salts.

CHAPTER 4 - GAS LAWS

Learning Focus

Relate pressure, volume and absolute temperature of gases, interpret graphs, solve gas-law calculations and connect particle behaviour to everyday and practical situations.

4.1 Gas Particles and Pressure

Gas pressure results from particle collisions with container walls.

4.2 Kelvin Temperature

 **REMEMBER:** Gas-law temperature ratios use Kelvin, not degrees Celsius.

 **COMMON ERROR:** Never substitute 27 directly into a gas-law temperature ratio; use 300 K.

$T(K) = t(^{\circ}C) + 273$. Gas-law equations involving absolute temperature require Kelvin.

4.3 Boyle's Law

At constant temperature for a fixed amount of gas, pressure is inversely proportional to volume:

$$P_1V_1 = P_2V_2$$

 **WORKED EXAMPLE:** A gas occupies 600 cm³ at 100 kPa and is compressed to 400 cm³. $P_2 = (100 \times 600)/400 = 150$ kPa.

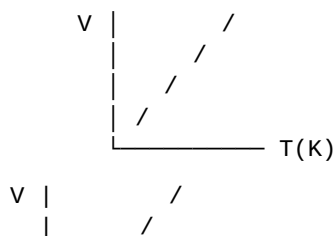
 **APPLICATION:** A syringe demonstrates how reducing gas volume increases pressure when temperature is approximately constant.



4.4 Charles's Law

At constant pressure, volume is directly proportional to absolute temperature: $V_1/T_1 = V_2/T_2$.

 **WORKED EXAMPLE:** 300 cm³ at 27°C = 300 K. At 127°C = 400 K: $V_2 = 300 \times 400/300 = 400$ cm³.



3. Describe preparation of 250 cm³ of 0.20 mol dm⁻³ NaOH, including calculation of mass and correct volumetric technique. [10]
4. Apply nitrogen/ammonia chemistry to fertiliser production, agriculture and eutrophication. [10]
5. Describe the sulfur-to-sulfuric-acid pathway, including equations, catalyst and environmental considerations. [7]
6. Identify ammonia from observations and chemical tests. [5]
7. Choose appropriate methods for preparing soluble and insoluble salts and write a precipitation ionic equation. [7]
8. A hydrated MgSO₄ sample changes from 2.46 g to 1.20 g on heating. Determine x in MgSO₄·xH₂O and explain constant mass. [8]
9. Apply Boyle's law to 480 cm³ at 120 kPa compressed to 320 cm³; explain the particle model. [7]
10. A balloon of 750 cm³ at 300 K cools to 240 K. Calculate its new volume and explain the change. [6]
11. Interpret reaction-rate data at different temperatures using average rate and collision theory. [9]
12. Explain catalysts, activation energy, final yield and one economic advantage. [9]
13. Classify C₂H₆, C₃H₆ and C₄H₁₀; state general formulae and bromine-water behaviour. [8]
14. Represent ethene addition with bromine and explain why it is an addition reaction. [5]
15. Explain hazards of incomplete combustion in a household setting and state safety measures. [7]
16. Design a fair investigation of particle size and the reaction between CaCO₃ and HCl. [14]
17. Integrated Industrial Chemistry: connect mole calculations, catalysts, combustion, emissions and decision-making. [11]

FINAL NATIONAL ASSESSMENT PRACTICE TEST 1**Grade 11 Senior School Chemistry | Time: 2 Hours | Total: 80 Marks**

Answer all questions. Show working in calculations and give units where required.

SECTION A — 20 MARKS

1. Calculate the amount, in moles, contained in 9.0 g of water. (H = 1, O = 16) [2]
2. A farmer applies a nitrogen fertiliser to maize. State one benefit and one possible environmental effect of excessive application. [2]
3. State two main steps used after reacting excess copper(II) oxide with dilute sulfuric acid to obtain copper(II) sulfate crystals. [2]
4. A fixed mass of gas is warmed in a flexible container. State the expected change in volume and name the gas law involved. [2]
5. State two factors that can increase the rate of a reaction. [2]
6. State one safety reason why incomplete combustion of a fuel in a poorly ventilated room is dangerous. [2]
7. Classify C_3H_8 as an alkane or alkene and state the general formula of its homologous series. [2]
8. State two products that may be formed during incomplete combustion of a hydrocarbon. [2]
9. State the reagent/ion used to test for chloride ions and the observation for a positive test. [2]
10. State two observations/tests that can be used to identify ammonia gas. [2]

SECTION B — 40 MARKS

11. A learner prepares 250 cm³ of 0.20 mol dm⁻³ sodium chloride solution. (a) Calculate the amount of NaCl required. (b) Calculate its mass, $M_r(\text{NaCl})=58.5$. (c) Describe how the solution is prepared accurately using a volumetric flask. [8]
12. Ammonia is manufactured from nitrogen and hydrogen. (a) Write the balanced reversible equation. (b) State the catalyst. (c) Explain why unreacted gases are recycled. (d) Explain one agricultural benefit and one environmental risk associated with nitrogen fertilisers. [8]
13. In an experiment, 28 cm³ of gas is collected in 20 s. (a) Calculate the average rate. (b) Explain, using collision theory, how increasing temperature changes the rate. (c) State two variables that should be controlled when comparing temperatures. [8]
14. A gas occupies 600 cm³ at 100 kPa. It is compressed at constant temperature to 400 cm³. (a) Calculate the new pressure. (b) Explain the pressure change using the particle model. (c) State the condition that must remain constant for Boyle's law. [8]
15. For each of the following, state a suitable salt-preparation method and outline the main steps: (a) copper(II) sulfate from copper(II) oxide and dilute sulfuric acid; (b) silver chloride from soluble solutions. Include the ionic equation for formation of AgCl. [8]

SECTION C — 20 MARKS

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